look considerably different.

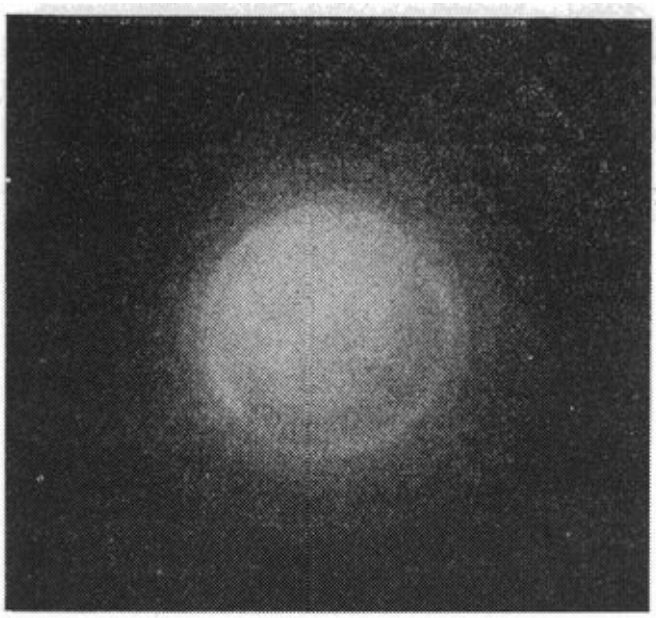


Figure 5.—Interference rings of Rëntgen rays.

Modern physics utilizes both the wave aspect and the particle aspect, and treats both as of equal right. For we know when to use one and when the other, and we know also that neither can stand by itself, without the other.



Figure 6.—Interference rings of Beta rays.

Thus we see that it is very difficult to obtain a visual concept of an atom and at this point, we have evidently reached the limit of the possibilities of visualization. As we have seen, the most that can be observed of an atom, is the result of one single snapshot. But such a snapshot never shows the orbit of the electron, but two specific points only—the nucleus and the electron in its momentary position. If a great number of such photographs are taken, in succession, of different atoms of the same kind, the electron will be found in various locations, at a greater or lesser distance from the nucleus, more frequently in one spot, less frequently in another. In this way, we eventually obtain a general, overall picture of the probability of finding the electron at